WORKED SOLUTION

STEP II 2008, Question 12

Probability and Statistics • Independent Events

STEP II 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q12

The key idea

The expectation of a count is the sum of the probabilities of the individual events that make it up, so the first part is one quadratic in p. For the second part, the judges toss their mental coins without looking at the defendant, so the King's pronouncement is independent of the truth. Writing the probability of a correct pronouncement in terms of t and then rearranging it around $\frac{1}{2}$ makes the answer fall out in one line.

Solution

Finding p

Let A, B and C be the events that the ass, the beaver and the centaur respectively give a guilty verdict, and let I_A, I_B, I_C be their indicator variables (equal to 1 if the event happens and 0 otherwise). Then $X = I_A + I_B + I_C$, and since $\mathrm{E}(I_A) = \mathrm{P}(A)$ and so on, linearity of expectation gives

$$\mathrm{E}(X) = p + \frac{p}{3} + p^2.$$

(Linearity doesn't need independence; we use independence only later.) Setting this equal to $\frac{4}{3}$ and multiplying by 3:

$$3p^2 + 4p - 4 = 0 \quad \Longrightarrow \quad (3p-2)(p+2) = 0.$$

A probability can't be negative, so $p + 2 \neq 0$, and therefore

$$p = \frac{2}{3}.$$

So the three judges say "guilty" with probabilities

$$\mathrm{P}(A) = \frac{2}{3}, \qquad \mathrm{P}(B) = \frac{2}{9}, \qquad \mathrm{P}(C) = \frac{4}{9},$$

all of which lie in $[0, 1]$, as they must.

The probability that the King says "guilty"

Let K be the event that at least two judges give a guilty verdict, so that K is the event that the King pronounces the defendant guilty. Write a, b, c for the three probabilities above. Exactly two guilty verdicts can happen in three ways and all three in one way; using the independence of the judges,

$$\begin{aligned} \mathrm{P}(K) &= ab(1-c) + ac(1-b) + bc(1-a) + abc \\ &= ab + ac + bc - 2abc. \end{aligned}$$

Here $ab = \frac{4}{27}$, $ac = \frac{8}{27}$, $bc = \frac{8}{81}$ and $abc = \frac{16}{243}$, so

$$\mathrm{P}(K) = \frac{36}{243} + \frac{72}{243} + \frac{24}{243} - \frac{32}{243} = \frac{100}{243}.$$

Call this g for short.

Finding t

Let D be the event that the defendant really is guilty, so $\mathrm{P}(D) = t$. The King is correct when D and K both happen, or when neither does.

Ask yourself Why can we multiply $\mathrm{P}(D)$ by $\mathrm{P}(K)$?

The judges decide "entirely at random": their verdicts do not depend on the evidence, so K is independent of D. Hence

$$\mathrm{P}(\text{correct}) = \mathrm{P}(D)\,\mathrm{P}(K) + \mathrm{P}(D')\,\mathrm{P}(K') = tg + (1-t)(1-g).$$

Now subtract $\frac{1}{2}$ and factorise:

$$\begin{aligned} tg + (1-t)(1-g) - \tfrac{1}{2} &= 2tg - t - g + \tfrac{1}{2} \\ &= (2t-1)(g - \tfrac{1}{2}). \end{aligned}$$

So the King is correct with probability $\frac{1}{2}$ exactly when $(2t-1)(g-\frac{1}{2}) = 0$. Since $g = \frac{100}{243} \neq \frac{1}{2}$, the second factor is non-zero, and we must have

$$t = \frac{1}{2}.$$

Equivalently, without the factorisation: $\frac{100}{243}t + \frac{143}{243}(1-t) = \frac{1}{2}$ gives $143 - 43t = \frac{243}{2}$, so $43t = \frac{43}{2}$ and $t = \frac{1}{2}$.

Notice that the value of p hardly matters here: whatever the judges' probabilities, provided $\mathrm{P}(K) \neq \frac{1}{2}$, a court that ignores the evidence is right half the time exactly when guilty and innocent defendants are equally common.

Checks

- With $p = \frac{2}{3}$: $\frac{2}{3} + \frac{2}{9} + \frac{4}{9} = \frac{12}{9} = \frac{4}{3}$, as required.

- Enumerating all eight verdict patterns exactly with fractions gives $\mathrm{P}(K) = \frac{100}{243} \approx 0.412$ and $\mathrm{E}(X) = \frac{4}{3}$.
- A simulation of 200 000 trials with $t = \frac{1}{2}$ gave the King correct in a proportion 0.5007 of cases.
- Sanity check of the formula: if $t = 1$ (everyone is guilty) the King is right with probability g, and if $t = 0$ with probability $1 - g$; the probability is linear in t, so it passes through $\frac{1}{2}$ at the midpoint.

Common slips

- Taking "at least two" as $ab + ac + bc$. This counts the all-three case three times; the correct expression subtracts $2abc$ (or add up the four disjoint cases directly).
- Forgetting to reject $p = -2$, or not checking that p^2 and $p/3$ are valid probabilities.
- Treating "the King pronounces correctly" as $\mathrm{P}(K)$ alone, forgetting the innocent defendants who are correctly acquitted.

Beyond STEP

The first part is a small instance of a powerful trick: writing a count as a sum of indicator variables, so that its expectation is just a sum of probabilities, even when the events are dependent. Undergraduate probability uses this constantly, for example to find the expected number of fixed points of a random permutation. The second part is about a classifier whose output is independent of the truth, which is the starting point for thinking about sensitivity, specificity and Bayes' theorem in testing. Blitzstein and Hwang's *Introduction to Probability* makes heavy use of indicator variables and is a good next step.

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